

Quick Check

Lesson 4: Following the Pollen Tube: How Double Fertilisation Explains Why a Lone Pawpaw Flowers but Sets No Fruit

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.1: Reproduction in Plants | Lesson 4

A pollen grain lands on the stigma of a compatible flower. Which route do the male nuclei take to reach the egg cell?

- A** The pollen grain mixes with nectar and flows down the stem to the ovule
- B** The pollen grain falls straight through the petals onto the egg cell
- C** The pollen grain stays on the stigma and the egg cell moves up the style to meet it
- D** The pollen grain germinates and a pollen tube grows down the style to the ovary, then into the ovule to the egg cell

A pollen tube has grown down the style and into the ovary and an ovule. Through which structure does it enter the ovule to reach the embryo sac?

- A** The filament
- B** The petal base
- C** The stigma
- D** The micropyle

When a pollen tube reaches the embryo sac, it releases two sperm nuclei. What does each sperm nucleus do?

- A** One sperm nucleus fuses with the tube nucleus, and the other fuses with the egg cell
- B** One sperm nucleus fuses with the egg cell, and the other disappears because it is not needed
- C** One sperm nucleus fuses with the egg cell, and the other fuses with the two polar nuclei
- D** Both sperm nuclei fuse with the egg cell to make a strong zygote

A learner says: "Pollination and fertilisation are the same thing." Which statement correctly corrects this learner?

- A** Pollination happens inside the ovule, and fertilisation happens on the stigma
- B** Pollination is the transfer of pollen to the stigma, and fertilisation is the fusion of the sperm nuclei with the egg cell and polar nuclei
- C** Pollination is the fusion of a sperm nucleus with an egg cell, and fertilisation is the landing of pollen on the stigma
- D** Pollination happens only in maize, and fertilisation happens only in beans

A single pawpaw tree in a compound in Kisumu is covered in flowers but sets no fruit. There is no other pawpaw tree nearby. Which explanation is best supported by what happens inside a flower?

- A** The tree has too many flowers, so the pollen is used up before it reaches the stigma
- B** The tree makes seeds without fertilisation, but they fall off before they can be seen
- C** The flowers need to be cut off so that the stem can grow into a new plant
- D** No compatible pollen arrives, so no pollen tube grows, no gametes fuse, and no zygote or endosperm forms, so there is no seed or fruit

In maize, the egg cell, each sperm nucleus and each polar nucleus have 10 chromosomes. One sperm nucleus fuses with the two polar nuclei to form the endosperm nucleus. How many chromosomes does the endosperm nucleus have?

A 40

B 30

C 20

D 10

A Kenyan farmer grows cassava from stem cuttings and maize from seeds. Which statement correctly contrasts the two methods?

- A** Cassava cuttings need a compatible partner plant for pollen, while maize seeds do not
- B** Maize grains form without fertilisation, while cassava cuttings form a zygote and an endosperm
- C** Cassava cuttings grow into new plants without any fusion of gametes, while a maize grain forms only after fertilisation of the egg cell and polar nuclei
- D** Both cassava cuttings and maize grains form only after the fusion of gametes

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).